Quiz: Memory

This quiz has 5 questions. The passing score is 4. Back Next

Started on	Tuesday, January 14, 2025, 8:04 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 8:06 PM
Time taken	2 mins 14 secs
Grade	**2.00** out of 5.00 (**40**%)

Question **1**

Incorrect

0.00 points out of 1.00

In z/Architecture, a byte's most significant bit is numbered 7 and is the rightmost

Select one:

- True ✖
- False

In z/Architecture, a byte's **most significant bit** is **numbered 0** and is the **leftmost**.

A byte's **least significant bit** is **numbered 7** and is the **righmost**.

Bits in a byte

Bit number	01234567
Contents (example)	11110001

The correct answer is 'False'.

Question **2**

Correct

1.00 points out of 1.00

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1,048,576 bytes are called a Megabyte.

Select one:

- True ✔
- False

Correct!

1,048,576 is 2^{20}. 1,048,576 bytes are called a Megabyte (abbreviated MB).

Similarly, 2^{10} bytes (1,024 bytes) are called a Kilobyte (KB), and 2^{30} bytes are called a Gigabyte (GB).

The correct answer is 'True'.

Question **3**

Incorrect

0.00 points out of 1.00

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In z/Architecture, a byte’s leftmost bit is the most significant and is numbered 0

Select one:

- True
- False ✖

In z/Architecture, bits in a byte are **numbered 0 to 7**, **left to right**, with **bit 0** being the **most significant**.

Bits in a byte

Bit number	01234567
Contents (example)	11110001

The correct answer is 'True'.

Question **4**

Incorrect

0.00 points out of 1.00

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The memory address 00A004 can refer to

Select one or more:

- [ ] A. An aligned doubleword
- [ ] B. An aligned quadword
- [ ] C. An aligned fullword
- [x] D. An aligned halfword ✔
- [ ] E. A byte

The memory address **00A004** is a **multiple of 4** (you can check by converting 00A004 to decimal, but you can tell by just looking at the address: hexadecimal numbers ending with **0**, **4**, **8**, or **C** are multiples of 4).

A multiple-of-4 address can refer to:

- a **byte** (because all addresses reference bytes)
- an **aligned halfword** (aligned halfwords must be on multiple-of 2 addresses, and a multiple of 4 is also a multiple of 2)
- an **aligned fullword** (aligned fullwords must be on a multiple-of-4 address)

The address **cannot refer to aligned doublewords or quadwords**, because **it is not a multiple of 8 or 16**.

The correct answers are: An aligned halfword, An aligned fullword, A byte

Question **5**

Correct

1.00 points out of 1.00

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In EBCDIC, hexadecimal F1 represents the character ‘1’

Select one:

- True ✔
- False

Correct!

In EBCDIC, the characters '0' to '9' are represented by hexadecimal 'F0' to 'F9':

EBCDIC numbers

Byte content	Character represented
F0	'0'
F1	**'1'**
F2	'2'
...	
F9	'9'

The correct answer is 'True'.